COMP396 Weekly Meeting Minutes

(Week 6)

Date & Time: Thursday, October 30th, 2025, 16:30-18:30

Location: Sydney Jones Library / Lower Ground Group Study Table 1

Note Taker: Zhao Saria

Attendees: Zhao Saria, Zhang Olivia, Xue Yudan, Wong Haw

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1. Agenda:

- Review the completion status of the previous week's (Week 5) tasks (E1-E3).
- Conduct a Code Review for Strategy S2 (Buy and Hold) and Strategy S3 (Z-score Mean Reversion).
- Report on the Peer Review results for the S1 strategy code.
- Report on the research progress of CVaR portfolio optimization and the implementation details for Part 2.
- Discuss and finalize the query for the Supervisor regarding the £1,000,000 starting capital allocation.
- Plan the core tasks for the next phase (Week 7), focusing on the combined strategy implementation and report writing.

2. Discussion & Decisions:

2.1. Task Review and Result Confirmation

Team members confirmed that all tasks assigned in the previous (Week 5) meeting have been completed on time:

- **Strategy Implementation (E1):** Zhao Saria has successfully completed the coding and initial testing for Strategy S2 (Buy and Hold) and Strategy S3 (Z-score Mean Reversion).
    - **Results:** S2-Annualized Sharpe ratio 1.335 and positive returns of 14.31%; S3-While absolute returns are modest (0.27%), the strategy achieves exceptional drawdown control (-0.37%) and maintains positive returns despite challenging testing conditions.

- **Code Review & Literature (E2):** Wong Haw has thoroughly read the reference material "Trading Systems" and conducted a detailed peer review of the S1 strategy code, confirming the robustness of the code logic.

- **Portfolio Research & Support (E3):** Xue Yudan and Zhang Olivia have researched literature on CVaR optimization strategies, further refining the implementation details for Part 2.

2.2. Individual Strategies Code-Complete

Team members conducted a collective review of the newly completed S2 and S3 strategy code from task E1. All three independent strategies (S1: Dual MA, S2: Buy & Hold, S3: Z-score) are now ready for integration into the Part 2 portfolio.

2.3. Capital Allocation Query (£1,000,000 Budget)

Team members discussed the ambiguity in the project brief regarding the application of the £1,000,000 starting capital across Part 1 (individual strategy testing) and Part 2 (combined portfolio). To ensure the accuracy of the report, we decided to pose this question to the Supervisor.

We summarized three possible interpretations:

- **Scenario 1:** In Part 1, each individual strategy (S1, S2, and S3) is backtested independently using £1,000,000 as its own starting capital for a fair, isolated

evaluation.

- **Scenario 2:** The three strategies in Part 1 must share the £1,000,000 capital (e.g., allocating £333k per strategy).
- **Scenario 3 (Our likely interpretation):** The individual strategy tests in Part 1 are just for validating the strategy logic (and the starting capital used here is just a baseline for calculating percentage returns). The total £1,000,000 capital is only applied in Part 2 to the final combined portfolio, where the CVaR optimizer will decide how to allocate this total among the three strategies.

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3. Task Allocation:

With the independent strategies completed, Team members' focus now officially shifts to implementing the combined strategy and drafting the Design Document.

ID	Module	Task Description	Owner	Due Date
F1	Portfolio Implementation	Complete the code for the **Combination Strategy (Part 2)**, integrating S1, S2, and S3, and applying the CVaR (SLSQP) optimizer for dynamic capital allocation. .	Zhao Saria	Nov. 5
		Complete the outline writing of the individual strategy section.	Zhao Saria	Nov. 3
F2	Report Writing (Part 1)	Complete the **Individual Strategy section of the Design Document**. This includes strategy type, data analysis, rationale, and	Zhang Olivia & Xue Yudan & Wong Haw	Nov. 5

		parameter optimization.		
F3	Report Writing (Part 3)	Summarizes the **specific individual contributions** to the group process and writes a first draft for the "Breakdown of Teamwork" section.	All Members	Nov. 8
F4	Project Management	Organize the three scenarios regarding the £1,000,000 capital allocation (Section 2.3) and formally **ask the Supervisor** for clarification in next week's meeting.	Zhang Olivia	Nov. 6

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4. Current Challenges and Issues

- **CVaR Optimization Implementation:** The primary technical challenge is now task F1. Specifically, how to accurately implement the CVaR (SLSQP) optimization algorithm within the Backtrader framework and ensure it converges stably and efficiently during the 20-day rebalancing period.
- **Report Writing:** Team members must transition from a pure coding phase to a high-quality academic writing phase, ensuring the descriptions in tasks F2 and F3 are clear, professional, and meet the requirements of the Design Document.

5. Supervisor Question

- The core question for this week (F4) is regarding the allocation

method for the £1,000,000 starting capital.

6. Next Meeting Arrangement

- **Date & Time:** Thursday, November 6th, 2025, 15:00-17:00
- **Location:** Sydney Jones Library
- **Key Agenda Items:**
  - Code Review of the Combination Strategy (F1).
  - Review the first drafts of the Design Document (Part 1) and individual contributions (Part 3) (F2, F3).
  - Discuss the Supervisor's response regarding the capital allocation query (F4) and finalize the backtesting methodology for the report.
  - Draft and review the Report Part 2 section, covering the CVaR strategy type, data analysis, rationale, and parameter optimization.
  - Confirm the modification plan for the strategies and report (currently based on Data 1) for when Data 2 is released.
  - Plan the final sprint tasks before the first report submission.